



Ionic liquids for the separation of benzene and cyclohexane – COSMO-RS screening and experimental validation

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ABSTRACT

The separation of benzene and cyclohexane from their mixture is difficult to perform via conventional distillation because of their close boiling points. In this work, liquid-liquid extraction using ionic liquids (ILs) is suggested for this purpose and 16 cations and 13 anions were selected to form 208 possible ILs screened with the Conductor-like Screening Model for Real Solvents (COSMO-RS) module. The screening result was experimentally validated by liquid-liquid extraction using four of the top ranked ILs, namely 1-ethyl-3-methylimidazolium acetate ([C₂mim][Ac]), 1-ethyl-3-methylimidazolium dicyanamide ([C₂mim][N(CN)₂]), 1-ethyl-3-methylimidazolium thiocyanate ([C₂mim][SCN]) and 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([C₂mim][Tf₂N]). The ternary liquid-liquid equilibria for these ILs with benzene and cyclohexane were investigated at 25 °C and 1 atm with feed concentration of benzene ranging from 10 to 60 wt%. Good agreement was achieved between the tie-lines obtained from the COSMO-RS model and those obtained experimentally. The performance of ILs used in this study was compared with organic solvents, other ILs, and deep eutectic solvents reported in literature. The results of selectivity and distribution ratio confirmed that COSMO-RS was a reliable method for solvent screening and demonstrated the suitability of the selected ILs as extracting solvents. In all ternary systems, no IL was detected in the cyclohexane layer and the concentration of cyclohexane in the IL layer was very low. This observation indicated that there was minimum cross-contamination between the phases and therefore less energy will be required for the solvent recovery.

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1. Introduction

Benzene and cyclohexane are two of the most valuable products that are widely processed in the petrochemical industry. At present, nearly all cyclohexane is produced by the catalytic hydrogenation of benzene [1]. In this process, the cyclohexane with high purity can only be produced via complex process control that involves a complex heat integration and economic study [2, 3]. Therefore, this method usually produces a mixture of benzene and cyclohexane. The unreacted benzene in the reactor's effluent must be removed to produce pure cyclohexane. However, the separation of benzene and cyclohexane by conventional distillation is quite difficult and challenging because they have close boiling points (80.1 °C for benzene and 80.7 °C for cyclohexane).

The conventional techniques employed for the separation of benzene and cyclohexane from their mixture include azeotropic distillation and extractive distillation [4]. In azeotropic distillation, a strongly polar entrainer is introduced to form an azeotropic mixture with cyclohexane, which alters the vapor-liquid equilibrium (VLE) curve for this mixture. Meanwhile, the extractive distillation method uses the entrainer to

reduce the volatility of benzene. Despite their potential at an industrial scale, both processes suffer from several disadvantages such as process complexity, high capital and operating costs, and high energy consumption [5, 6]. Moreover, these processes are carried out by adding a third compound as the entrainer; the removal of the third compound from the distillate introduces higher process complexity and cost to the adopted method. In addition, azeotropic distillation and extractive distillation are only suitable for high (>90%) and medium (65–90%) concentration of benzene, respectively. Liquid-liquid extraction (LLE) is widely used in the industry because of the mild operating conditions and simplicity. Although LLE is suitable for separating low concentrations of benzene (20–65%), there is still no industrial process available for concentrations <20 wt%. While the pervaporation process was reviewed as a potential technique, it was limited by the challenges encountered during economic evaluation [5]. Several reports have demonstrated the excellent performance of industrial organic solvents such as ethylene glycol, tetra-ethylene glycol, sulfolane, and *N*-methylpyrrolidone as extractants [7–9]. However, their application is associated with several challenges because these solvents are usually volatile, toxic and flammable.

Interestingly, ionic liquids (ILs) have become a progressively popular class of solvent in the last decades as their potential in many industrial

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applications has become more diverse. Compared to other solvents, ILs are exceptionally non-flammable, non-volatile, environmentally benign, and thermally stable over a wide range of temperature. ILs have also been recognized for their remarkable ability to break the azeotropic properties in many mixtures such as ethanol-water, alcohol-alkane, aromatic-aliphatic, ketone-alkane, etc. [10]. The reports of new data for ternary phase equilibria between several ILs, benzene, and cyclohexane proved the feasibility of using ILs as an extracting solvent [11–14].

The necessity of developing a screening process is based on the idea that many other ILs having high extracting performances might remain undiscovered. This is because of the availability of a large number of available cations and anions, which makes the selection of potential ILs through experimental work a time-consuming and expensive process. In order to develop an effective selection procedure, the potential ILs should undergo a preliminary screening. Therefore, the use of reliable predicting tools is crucial. In this regard, a computational program known as the conductor-like screening model (COSMO-RS) has attracted significant attention and gained influence in research activity because of its ability to describe the thermodynamic properties and behavior of ILs. By combining the statistical thermodynamics and quantum chemical calculations, COSMO-RS can determine and predict the thermodynamic properties of mixtures without requiring any experimental data.

For the separation of benzene and cyclohexane by liquid-liquid extraction, COSMO-RS screening was previously performed by Lyu et al. using 12 cations and 22 anions, which found 1-butyl-3-methylimidazolium tetrachloroaluminate, $[C_4mim][AlCl_4]$ as the best IL [15]. In their study, the new $[AlCl_4]^-$ anion was introduced using quantum chemical calculations and included into the anion list. Gonfaa et al. used COSMO-RS to perform a preliminary screening of various ILs and utilized nitrile functionalized ILs to separate benzene and cyclohexane by the extractive distillation process [16]. Recently, deep eutectic solvent (DES), which is often regarded as an analogue of ILs was utilized as a solvent for this process. Some DESs have been screened using COSMO-RS for the separation of benzene and cyclohexane, [17] and other aromatic-aliphatic mixtures [18].

In this work, we disregarded the pre-determined ions or functional groups, and adopted a basic approach by choosing only the common cations and anions. This will provide a baseline for an advanced selection of ILs as the screened ILs would be potentially available in the market, and less complex or easy to synthesize. Furthermore, the nature of the selected ILs in this study can be developed further to enhance their extractive performance such as by modifying the structure [19], developing a binary IL mixture [20], or using co-solvents [21].

2. Computational methods

The computational calculations were carried out using the COSMOtherm software with BP-TZVP C30_1401 parameterization. The data prediction was obtained through density functional theory (DFT) calculations, utilizing the Becke-Perdew (BP) functional with the resolution of identity (RI) approximation and the Triple- ζ Zeta Valence Potential (TZVP) basis set. For the selection of cation, four types of cyclic cations were chosen: imidazolium, pyridinium, piperidinium, and pyrrolidinium. As shown in Table 1, each of these cations was expanded into four kinds of methyl-alkyl groups to study the effect of cation alkyl length. Meanwhile, for the selection of anion, thirteen common and less complicated anions were shortlisted, as shown in Table 2.

2.1. Prediction theory

For predictions involving ILs, the electroneutral approach was adopted where the IL is assumed to undergo complete dissociation into its respective cation and anion. Therefore, as shown in Eq. 1, the sigma profile of the IL (P_{IL}) was the linear addition of the sigma profiles

of both cation (P_{cation}) and anion (P_{anion}), such that the profile of the IL behaved as the profile of a single molecule.

$$P_{IL}(\sigma) = P_{anion}(\delta) + P_{cation}(\delta) \quad (1)$$

2.2. Calculation methodology

The activity coefficient at infinite dilution (γ) was obtained from the Chemical Potential menu in the COSMO-RS software, where the mole fractions of benzene and cyclohexane were set to zero. The methods and calculations required to obtain activity coefficient at infinite dilution has been described in detail by Klamt [22]. At infinite dilution, the capacity (C^∞) and selectivity (S^∞) of the IL are given by Eqs. 2 and 3, respectively.

$$C^\infty = \frac{1}{\gamma_1^\infty} \quad (2)$$

$$S^\infty = \frac{\gamma_2^\infty}{\gamma_1^\infty} \quad (3)$$

where γ_1^∞ and γ_2^∞ are the activity coefficients at infinite dilution for benzene and cyclohexane, respectively. The performance index at infinite dilution (PI^∞) of each IL was calculated as a product of capacity and selectivity at infinite dilution using Eq. 4:

$$PI^\infty = C^\infty \times S^\infty \quad (4)$$

2.3. Chemicals and materials

The details of the chemicals used in this work are summarized in Table 3 where all of them were used as received without further purification. For the determination of the ternary composition in the extract and raffinate phase at equilibrium, Nuclear Magnetic Resonance (NMR) spectroscopy was performed using either deuterated chloroform ($\geq 99.8\%$, stabilized with silver) or deuterated methanol (99.8%).

Table 1
List of cation in the COSMO-RS screening.

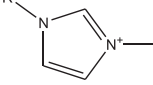
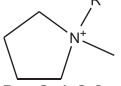
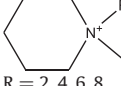
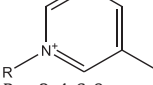
No	Cation name	Abbreviation	Chemical structure
1	1-alkyl-3-methylimidazolium	$[C_nmim]^+$	
2	1-alkyl-1-methylpyrrolidinium	$[C_nmpyrro]^+$	
3	1-alkyl-1-methylpiperidinium	$[C_nmpip]^+$	
4	1-alkyl-3-methylpyridinium	$[C_nmpyr]^+$	

Table 2

List of anion in the COSMO-RS screening.

No	Name	Abbreviation	Chemical structure
1	Acetate	[Ac] [−]	
2	Benzoate	[BzO] [−]	
3	Tetrafluoroborate	[BF ₄] [−]	
4	Dicyanamide	[N(CN) ₂] [−]	
5	Ethylsulfate	[EtSO ₄] [−]	
6	Hydrogensulfate	[HSO ₄] [−]	
7	Nitrate	[NO ₃] [−]	
8	Octylsulfate	[OcSO ₄] [−]	
9	Hexafluorophosphate	[PF ₆] [−]	
10	Salicylate	[Sal] [−]	
11	Bis(trifluoromethylsulfonyl)imide	[Tf ₂ N] [−]	
12	Thiocyanate	[SCN] [−]	
13	Methanesulfonate	[MeSO ₃] [−]	

Table 3

Chemicals used in this work.

Chemical name	CAS no.	Supplier	Purity (mass %)	Density (g·mL ^{−1})
Benzene	71-43-2	Merck	≥ 99.7	0.88
Cyclohexane	110-82-7	Merck	≥ 99.5	0.78
1-ethyl-3-methylimidazolium acetate	143314-17-14	Sigma Aldrich	≥ 95.0	1.027
1-ethyl-3-methylimidazolium dicyanamide	370865-89-7	Sigma Aldrich	98.0	1.06
1-ethyl-3-methylimidazolium thiocyanate	331717-63-6	Merck	≥ 98.0	1.11
1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide	174899-82	Sigma Aldrich	≥ 98.0	1.06

3. Experimental

3.1. Liquid–liquid extraction experiment

The feed mixture containing 10 wt% benzene in cyclohexane was first prepared inside a 20-mL screw-capped scintillation vial using an analytical balance with accuracy of ± 0.0001 g. This procedure was repeated for other concentrations of benzene (i.e., 20, 30, 40, 50, and 60 wt%) in the feed. Based on the binary composition, the total weight of the feed mixture was fixed at 2 g. The IL was then mixed into every feed mixture in a 1:1 weight ratio. Parafilm was used to seal each vial and avoid component loss due to evaporation. The vials were then placed in an incubation shaker at 25 °C and 1 atm, where spring clamps were used to hold them in place while shaking. They were shaken at 200 rpm for 6 h. After stopping the mixing procedure, the mixture was left for undisturbed for 12 h to reach equilibrium. This period was deemed to be sufficient based on a settling time study conducted to ensure that the equilibrium state was fully reached.

3.2. Determination of the ternary molar composition

For compositional analysis, a drop of the sample (± 0.035 mL) was taken out from the extract and raffinate layers using a micropipette. For the extract layer, a bubble was purged from the micropipette tip to avoid cyclohexane contamination from the raffinate layer. This drop of sample was then dissolved in ± 0.7 mL of deuterated solvent placed inside an NMR tube. The sample and solvent in the tube were shaken carefully to form a homogenous mixture. To ensure homogeneity, the samples were dissolved in different solvents. Deuterated chloroform was used to dissolve all samples in the top layer and the bottom layer of the $[C_2mim][Ac]$ system. On the other hand, the samples from the bottom layer of the $[C_2mim][N(CN)_2]$, $[C_2mim][SCN]$, and $[C_2mim][Tf_2N]$ systems were dissolved in deuterated methanol. Each NMR tube was tightly sealed with parafilm to avoid chemical loss. NMR 400 MHz Bruker spectrometer was used to obtain the 1H NMR spectra of each component. The selected hydrogen peaks (chemical shifts in ppm) for calculation of the ternary composition are: benzene at ± 7.35 (6H), cyclohexane at ± 1.47 (12H), $[C_2mim][Ac]$ (3H) at 1.91, $[C_2mim][SCN]$ (3H) at ± 1.58 , $[C_2mim][N(CN)_2]$ (3H) at ± 1.56 and $[C_2mim][Tf_2N]$ (3H) at 3.85. Eq. 5 was used to calculate the molar fraction of each component in both layers.

$$x_i = \frac{H_i}{\sum_{i=1}^3 H_i} \quad (5)$$

where x_i is the mole fraction of species i and H_i is the hydrogen peak area in the component i .

The NMR analysis is one among the most accurate techniques and for all experiments, the average uncertainty on the mole fraction is estimated to be <0.003 .

3.3. Experimental selectivity and distribution ratio

The distribution ratio of benzene (D_B) and the solvent selectivity (S) of each IL were used to evaluate the extraction efficiency by using Eqs. 6 and 7.

$$D_B = \frac{x_B^1}{x_B^2} \quad (6)$$

$$S = \frac{x_B^1/x_C^1}{x_B^2/x_C^2} \quad (7)$$

where x_B and x_C are the concentrations of benzene and cyclohexane, respectively. The superscripts 1 and 2 represent the extract and raffinate phase, respectively.

3.4. Consistency tests

For the experimental results, the consistency tests were conducted using the Hand and Othmer–Tobias correlations. These two correlations were used to obtain the Hand [23] and Othmer–Tobias [24] equations, respectively.

$$\ln \left(\frac{x_{Bz}''}{x_{Cyc}''} \right) = a + b \ln \left(\frac{x_{Bz}'}{x_{IL}'} \right) \quad (8)$$

$$\ln \left(\frac{1-x_{Cyc}'}{x_{Cyc}'} \right) = c + d \ln \left(\frac{1-x_{IL}'}{x_{IL}'} \right) \quad (9)$$

where x_{Cyc} , x_{Bz} , and x_{IL} represent the concentrations of cyclohexane, benzene, and IL, respectively. The superscripts ' and '' refer to the extract and raffinate phases, respectively. The linearity of each plot (i.e. the value of R^2 close to unity) indicates the excellent degree of consistency for the ternary LLE tie lines reported in this work.

4. Results and discussions

4.1. Sigma profile of industrial solvents and ILs

Before selecting the types of cation and anion, the interactions between benzene, cyclohexane and ILs or organic solvents were firstly analyzed by sigma (σ) profile analysis. Thus, four organic solvents commonly used for the separation of aromatic-aliphatic mixtures, namely, sulfolane, *N*-methyl-2-pyrrolidinone, ethylene glycol, and tetraethylene glycol, were selected as the benchmarks. The σ -profiles of the industrial solvents and ILs with cyclic cations are illustrated in Fig. 1 and Fig. 2, respectively. This profile can be categorized into three regions: positive polarity ($\sigma < -0.0084$ eÅ²), non-polar (-0.0084 eÅ² $\leq \sigma \leq 0.0084$ eÅ²), and negative polarity ($\sigma > 0.0084$ eÅ²). The y-axis represents the surface area of the respective sigma regions. The regions in the positive and negative polarities can be viewed as the regions of the hydrogen bond donor and acceptor, respectively. The interactions between the molecules of a system are viewed in terms of a peak area comparison among the curves at negative and positive σ values.

As seen in sigma profile graphs, both benzene and cyclohexane have curves in the non-polar region. Therefore, to establish a good interaction with benzene, the extracting solvent should possess a non-polar character as well. Based on Fig. 1, the selected industrial solvents are capable of interacting with benzene as they show curves with a non-polar character with peaks at approximately -0.005 eÅ².

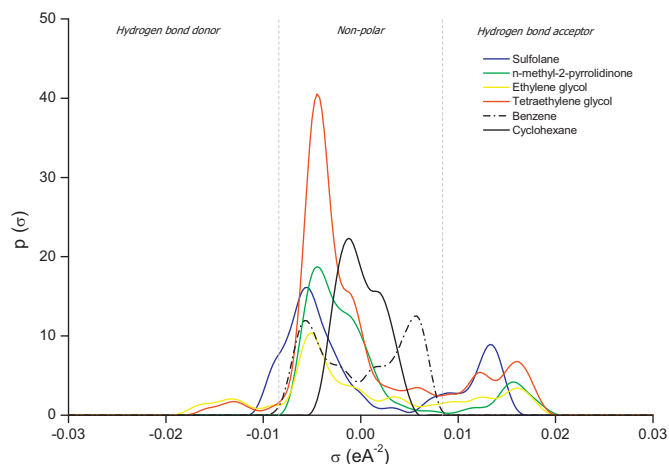


Fig. 1. Sigma profile of industrial organic solvents.

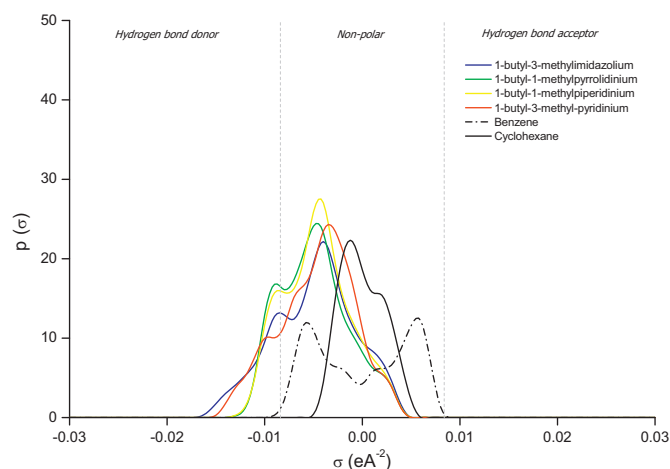


Fig. 2. Sigma profile of ILs with cyclic cations.

The increase in the alkyl chain from ethylene glycol to tetraethylene glycol led to a higher peak and created a larger interacting area with cyclohexane. In addition to its high interaction with benzene, tetraethylene glycol also exhibits high interaction with cyclohexane. Hence, tetraethylene glycol has a higher capacity but lower selectivity compared to ethylene glycol.

To analyze the interactions in the non-polar region, the σ -profiles of the ILs (Fig. 2) are represented by cations with same alkyl chains (butyl and methyl). In this work, the screening of cation is narrowed down to the cyclic types as the non-cyclic cations exhibited low performance in the extraction of aromatic compounds [25]. Similar to organic solvents, the major peak for IL cations also existed within the non-polar region. However, these cations also have curves at the polar regions, which indicated that they could be utilized for the extraction of both polar and non-polar solutes. In the non-polar region, the curves were of similar types and it was difficult to select the suitable cations using only the qualitative σ -profile. Thus, a quantitative screening was carried out in COSMO-RS by calculating the activity coefficients at infinite dilution to obtain the value of C^∞ , S^∞ and PI^∞ .

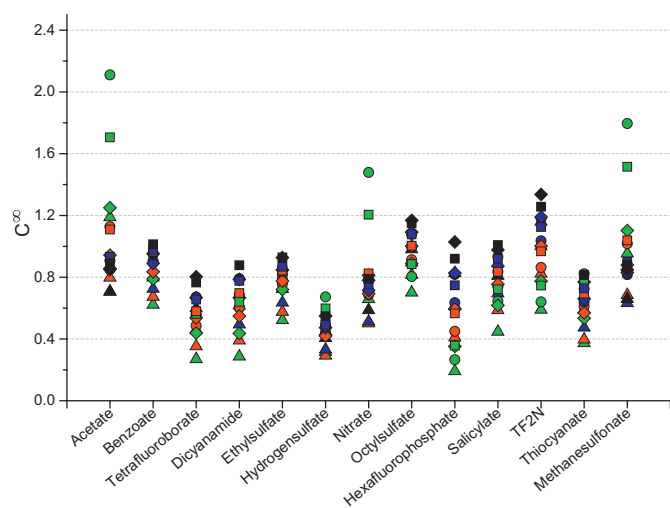


Fig. 3. Capacity of screened ILs at infinite dilution: $\blacktriangle$, $[C_n\text{mim}]^+$; $\blacklozenge$, $[C_n\text{mpyr}]^+$; $\bullet$, $[C_n\text{mpyrro}]^+$; $\blacksquare$, $[C_n\text{mpip}]^+$ and the fill colors of green, red, blue and black indicate the cation alkyl length (n) from ethyl, butyl, hexyl and octyl, respectively.

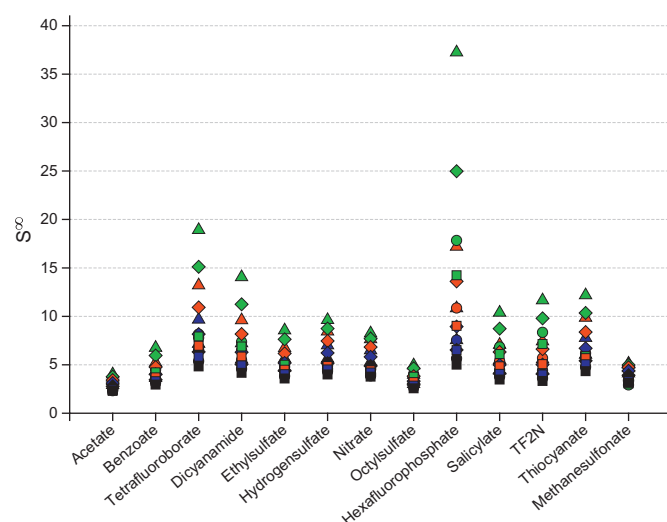


Fig. 4. Selectivity of screened ILs at infinite dilution: $\blacktriangle$, $[C_n\text{mim}]^+$; $\blacklozenge$, $[C_n\text{mpyr}]^+$; $\bullet$, $[C_n\text{mpyrro}]^+$; $\blacksquare$, $[C_n\text{mpip}]^+$ and the fill colors of green, red, blue and black indicate the cation alkyl length (n) from ethyl, butyl, hexyl and octyl, respectively.

4.2. Screening result

4.2.1. Capacity, selectivity and performance index

In this study, COSMO-RS was used to screen the suitable ILs by calculating their capacity, selectivity, and performance index at infinite dilution. As shown in Fig. 3, the capacity of the ILs generally increases with longer cation alkyl chain. As an example, for the IL series $[C_n\text{mim}][\text{EtSO}_4]$ ($n = 2, 4, 6, 8$), the capacity increased in the order of $[C_2\text{mim}][\text{EtSO}_4] < [C_4\text{mim}][\text{EtSO}_4] < [C_6\text{mim}][\text{EtSO}_4] < [C_8\text{mim}][\text{EtSO}_4]$. This could also be evaluated by the change in color for each IL symbol from white (ethyl group) to black (octyl group). However, a few ILs exceptionally showed an arbitrary relation with the cation alkyl chain length. This indicates that according to the COSMO-RS calculations, the capacity of the ILs in the extraction process may vary independently of their structural trends. In contrast, the selectivity of the ILs decreased with

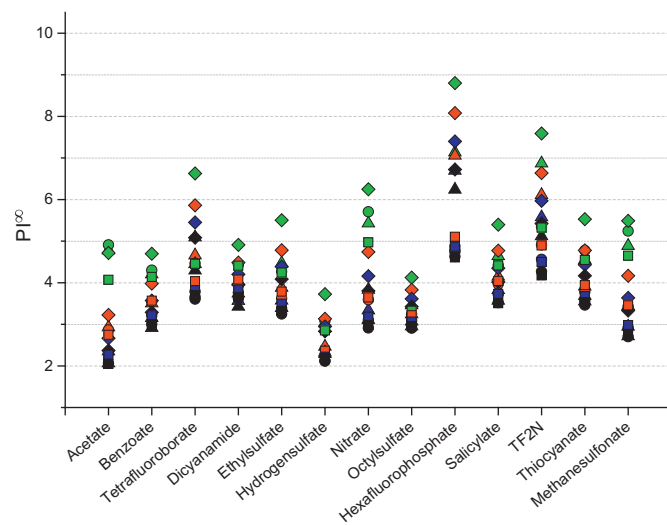


Fig. 5. Performance index of screened ILs at infinite dilution: $\blacktriangle$, $[C_n\text{mim}]^+$; $\blacklozenge$, $[C_n\text{mpyr}]^+$; $\bullet$, $[C_n\text{mpyrro}]^+$; $\blacksquare$, $[C_n\text{mpip}]^+$ and the fill colors of green, red, blue and black indicate the cation alkyl length (n) from ethyl, butyl, hexyl and octyl, respectively.

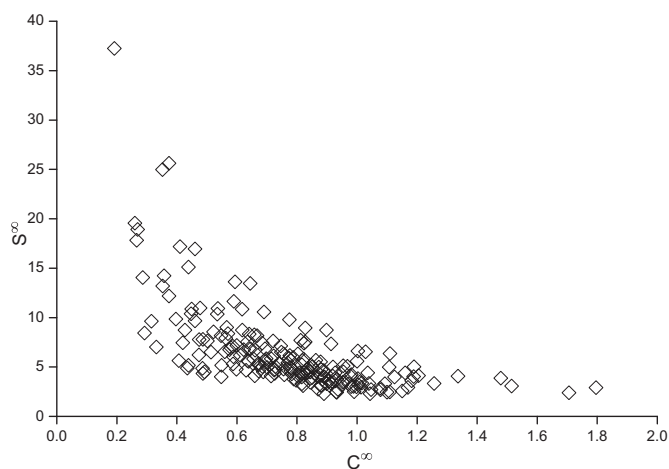


Fig. 6. S^{∞} vs C^{∞} for the screened ILs.

increasing cation alkyl length, as seen in Fig. 4. The capacity of an IL reflects its ability to extract other components regardless of their specific structure, while the selectivity evaluates the efficiency of the solvent for extracting a particular component, or more specifically, the solute (benzene in this case). The performance index in Fig. 5 combines both selectivity and capacity by a simple multiplication. Thus, the PI is a mathematical operation that enables evaluation of the overall extraction performance by accommodating the inverse relation between C^{∞} and S^{∞} . In liquid–liquid extraction process, capacity indicates the amount of solvent required, whereas selectivity evaluates the extraction efficiency. It is ideal to have an extracting solvent with high capacity and selectivity. However, the capacity of a solvent is generally inversely related to its selectivity, as seen in Fig. 6 for the selected ILs in this study. This relation makes it difficult to obtain ILs with high values of both capacity and selectivity. Therefore, in the solvent-screening process, a conflict exists for either one of these properties.

4.2.2. Selection of ILs for experimental validation

From the screening result, fifteen ILs for each evaluation criteria were shortlisted in Table 4. It is worth noting that the pyrrolidinium- and piperidinium-based ILs showed high C^{∞} , while the imidazolium- and pyridinium-based ILs showed high S^{∞} . This trend validates the effect of aromaticity of the cation toward the capacity and selectivity of ILs. Since benzene is an aromatic compound, the presence of C=C in the ring structure of imidazolium and pyridinium cations enhances the interaction between the cation and benzene via π - π interaction, thus increasing the selectivity. In contrast, the C=C bond is not present in

Table 4
The top 15 ILs according to COSMO-RS screening by C^{∞} , S^{∞} and PI.

No	Capacity		Selectivity		Performance index	
	IL	C^{∞}	IL	S^{∞}	IL	PI $^{\infty}$
1	[C ₂ mpyrro][Ac]	2.11	[C ₂ mim][PF ₆]	37.25	[C ₂ mpyr][PF ₆]	8.80
2	[C ₂ mpyrro][MeSO ₃]	1.80	[C ₂ mpyr][PF ₆]	24.98	[C ₄ mpyr][PF ₆]	8.08
3	[C ₂ mpip][Ac]	1.71	[C ₂ mim][BF ₄]	18.92	[C ₂ mpyr][Tf ₂ N]	7.59
4	[C ₂ mpip][MeSO ₃]	1.52	[C ₂ mpyrro][PF ₆]	17.83	[C ₆ mpyr][PF ₆]	7.40
5	[C ₂ mpyrro][NO ₃]	1.48	[C ₄ mim][PF ₆]	17.19	[C ₂ mim][PF ₆]	7.13
6	[C ₈ mpyr][Tf ₂ N]	1.34	[C ₂ mpyr][BF ₄]	15.12	[C ₄ mim][PF ₆]	7.06
7	[C ₈ mpip][Tf ₂ N]	1.26	[C ₂ mpip][PF ₆]	14.24	[C ₂ mim][Tf ₂ N]	6.87
8	[C ₂ mpyr][Ac]	1.25	[C ₂ mim][N(CN) ₂]	14.05	[C ₈ mpyr][PF ₆]	6.72
9	[C ₂ mpip][NO ₃]	1.20	[C ₄ mpyr][PF ₆]	13.61	[C ₆ mim][PF ₆]	6.69
10	[C ₆ mpyr][Tf ₂ N]	1.19	[C ₄ mim][BF ₄]	13.20	[C ₄ mpyr][Tf ₂ N]	6.64
11	[C ₂ mim][Ac]	1.19	[C ₂ mim][SCN]	12.18	[C ₂ mpyr][BF ₄]	6.63
12	[C ₈ mpyrro][Tf ₂ N]	1.18	[C ₂ mim][Tf ₂ N]	11.65	[C ₂ mpyr][NO ₃]	6.25
13	[C ₈ mpyr][OCsO ₄]	1.17	[C ₂ mpyr][N(CN) ₂]	11.25	[C ₈ mim][PF ₆]	6.24
14	[C ₈ mim][Tf ₂ N]	1.16	[C ₄ mpyr][BF ₄]	10.92	[C ₄ mim][Tf ₂ N]	6.11
15	[C ₈ mpip][OCsO ₄]	1.15	[C ₄ mpyrro][PF ₆]	10.86	[C ₆ mpyr][Tf ₂ N]	5.97

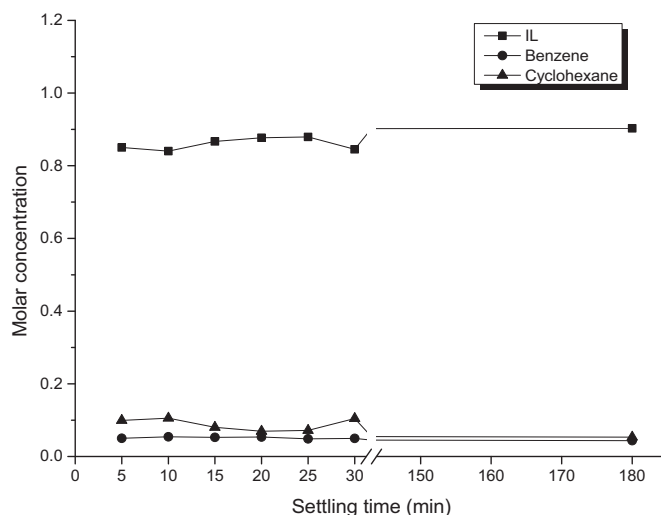


Fig. 7. Molar concentration of individual species in extract phase versus time after extraction.

the pyrrolidinium and piperidinium structure. This reduces the cation-benzene interactions, thus resulting in an increase in the capacity of the IL. The high value of S^{∞} for [C₂mim][SCN] observed in this work is consistent with a previous report wherein a different screening technique (gas liquid chromatography (GLC)) was utilized [26]. Since the COSMO-RS screening is an a-priori method which does not require experimental data, this result proved the reliability and advantage of using COSMO-RS to screen a large number of solvents.

It is known that despite the large number of possible ILs, only some are available in the market or can be possibly produced by laboratory synthesis. Our survey concluded that most of the ILs in Table 4 are not available commercially. Although ILs [C₂mim][PF₆], [C₄mim][PF₆], [C₂mpyrro][PF₆], [C₄mpyrro][PF₆] and [C₂mpyrro][Tf₂N] are commercially available, their use is economically limited owing to the extremely high cost. Despite their unavailability, the screening results could still be regarded as a motivation for their synthesis in the future.

Interestingly, four ILs in Table 4 are available for purchase at relatively low prices, namely, the high C^{∞} IL [C₂mim][Ac], the high S^{∞} ILs [C₂mim][N(CN)₂] and [C₂mim][SCN], and the high PI $^{\infty}$ IL [C₂mim]

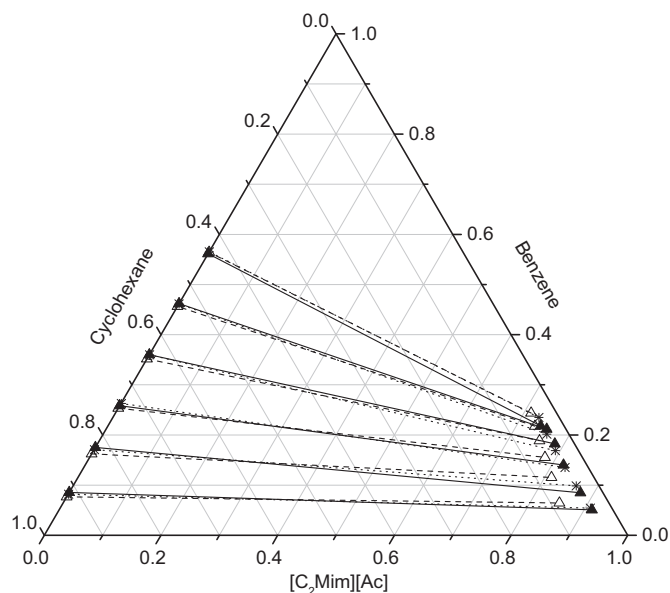


Fig. 8. Ternary LLE diagram for [C₂mim][Ac] + benzene + cyclohexane at 25 °C and 1 atm: —▲—, experimental; - - -, COSMO-RS; · · · · ·, NRTL.

[Tf₂N]. In addition, these ILs have also been recognized in other studies for different applications. [C₂mim][Ac] was found as a good IL for the dissolution of soft and hard wood [27] and for an effective conversion of cellulose into cellulose acetate [28]. [C₂mim][SCN] was reportedly highly selective in the desulfurization process [29] and was successfully utilized for the separation of aromatic/aliphatic mixtures [26]. The IL [C₂mim][N(CN)₂], apart from having low viscosity and water-miscible properties [30], was shown to be also capable of effectively extracting the heterocyclic sulfur compounds (thiophene and dibenzothiophene) from model oil [31]. It has been demonstrated that [C₂mim][Tf₂N] has an extremely high capacity for dissolving CO₂ gas [32] and shows potential in surfactant technology because of its aggregation behavior with some common non-ionic surfactants [33]. Considering the advantages of ILs over conventional solvents, the selection of these ILs for the separation of benzene and cyclohexane could be more economical and efficient.

4.3. Settling time study

Settling time is one of the requirements to ensure an optimal extraction process. In general, the equilibrium condition is characterized by equal values of pressure, temperature, and chemical potential in both phases. This study was performed to determine the minimum time for the top and bottom phases to reach equilibrium after the mixing is stopped. This could be represented by the mean of unchanged concentration of benzene in the extract layer. The ternary system of [C₂mim][Ac] + benzene + cyclohexane was taken as an example. As depicted in Fig. 7, although the mixture was left undisturbed for 3 h, the molar concentration of benzene in the extract layer remained constant, or reached equilibrium after 5 mins. The molar concentration of the IL and cyclohexane also remained almost constant throughout this time. The same settling results were observed for [C₂mim][N(CN)₂], [C₂mim][SCN], and [C₂mim][Tf₂N], indicating a good separation between the two phases after extraction.

4.4. Ternary LLE

The LLE diagrams for the ternary systems of [C₂mim][Ac]/[C₂mim][N(CN)₂]/[C₂mim][SCN]/[C₂mim][Tf₂N] + benzene + cyclohexane are displayed in Figs. 8–11, respectively. The molar compositions of the tie-lines for each ternary system are tabulated in Table 5. In addition,

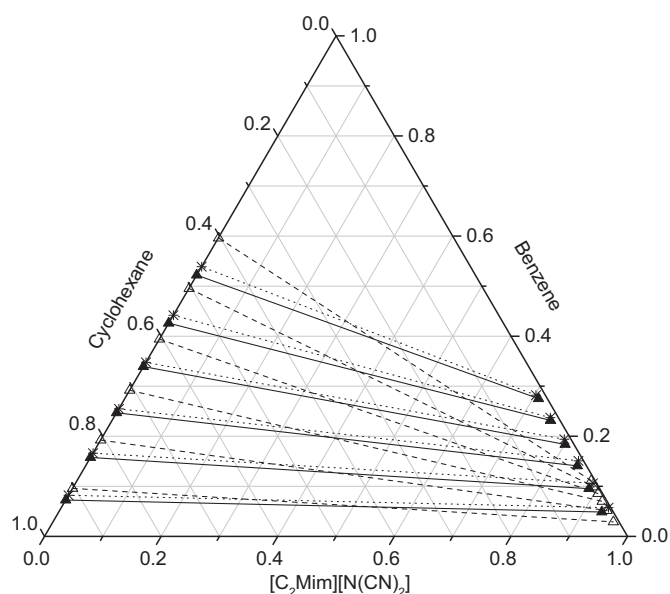


Fig. 9. Ternary LLE diagram for [C₂mim][N(CN)₂] + benzene + cyclohexane at 25 °C and 1 atm: ▲, experimental; –△–, COSMO-RS; ···, NRTL.

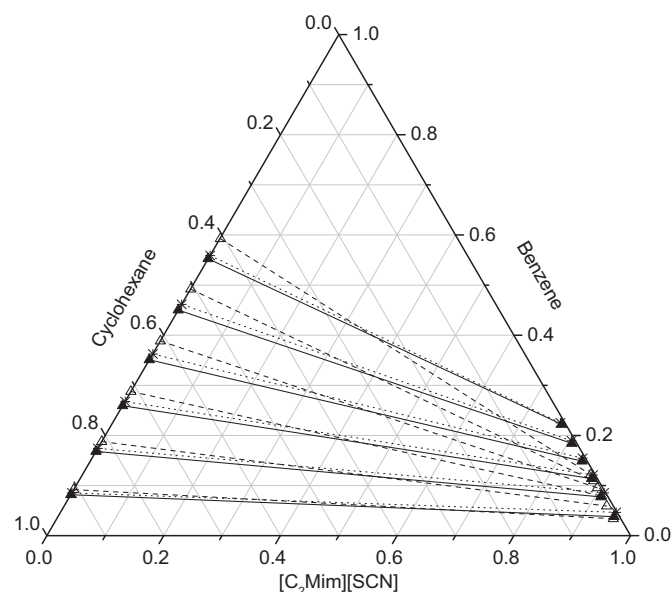


Fig. 10. Ternary LLE diagram for [C₂mim][SCN] + benzene + cyclohexane at 25 °C and 1 atm: ▲, experimental; –△–, COSMO-RS; ···, NRTL.

the value of parameters a , b , c , d , and R^2 for the consistency tests of each ternary system are listed in Table 6. As seen in these figures, the tie lines obtained from experimental and COSMO-RS approaches showed good agreement for all ternary systems, with the exception of [C₂mim][N(CN)₂] and [C₂mim][Tf₂N], which showed slight deviations. Moreover, COSMO-RS and experimental data also showed perfect agreement for the absence of ILs in the raffinate phase. The tie lines for every ternary system were negative, except in the case of [C₂mim][Tf₂N], where a positive slope was observed at higher concentrations of benzene. The absence of ILs in the cyclohexane-rich layer in all ternary systems indicates a favorable extraction process because the solvent cross-contamination could be avoided. In addition, the concentration of cyclohexane in the bottom layer was also very low, which indicated an easier regeneration of the IL.

4.5. COSMO-RS versus experimental results

As seen in Fig. 8, the COSMO-RS and experimental data for the system involving [C₂mim][Ac] achieved a very good agreement, especially in the raffinate phase. It is worth noting that although the predicted concentration of cyclohexane in the extract decreases steadily with increasing concentration of benzene, the experimental data remains almost constant. Thus, based on the experimental data, the low mutual solubility of cyclohexane with the ILs is constant regardless the concentration of benzene in the feed. This indicates a steady performance of ILs in retaining the low interaction with the cyclohexane, and therefore reduces the cross-contamination effect. In contrast to [C₂mim][Ac], the other three ILs showed close values between COSMO-RS and experimental approaches for the concentration of cyclohexane in the extract phase. COSMO-RS also estimated a higher slope than the experimental values for the tie lines in the ternary systems involving [C₂mim][N(CN)₂] (Fig. 9). This suggests that COSMO-RS underestimates the affinity of benzene toward the IL and overestimates its affinity toward cyclohexane. The ternary system involving [C₂mim][SCN] (Fig. 10) also showed similar results but with lower discrepancies. On the other hand, the predicted tie lines for the ternary system with [C₂mim][Tf₂N] showed negative slope with high magnitude, while the slope of the experimental tie lines were initially positive and changed to negative at higher feed concentrations of benzene (Fig. 11). This implies that the actual extraction of benzene using [C₂mim][Tf₂N] at lower feed concentration will produce distribution ratio of greater than unity.

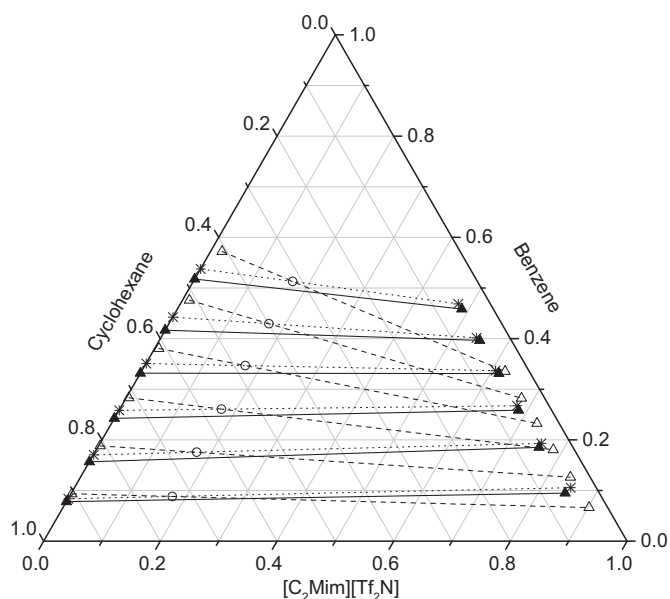


Fig. 11. Ternary LLE diagram for [C₂mim][Tf₂N] + benzene + cyclohexane at 25 °C and 1 atm: ▲, experimental; –△–, COSMO-RS; ·····, NRTL.

As seen in Fig. 12, the IL with a high Pf^* showed a high distribution ratio. Meanwhile, both COSMO-RS and experimental data gave the same trend of selectivity, [C₂mim][Ac] < [C₂mim][Tf₂N] < [C₂mim][N(CN)₂] < [C₂mim][SCN] (Fig. 13). Although the experimentally observed selectivity had a higher difference between [C₂mim][N(CN)₂] and [C₂mim][SCN] than predicted, it still proves that COSMO-RS is a reliable tool to find ILs with high selectivity.

The quantitative comparison between the experimental and COSMO-RS results can be evaluated from their value of RMSD. The

Table 5

Molar composition of the tie-lines with the distribution ratio and selectivity data for benzene (1) + cyclohexane (2) + ILs (3) at 25 °C and 1 atm.

Top layer			Bottom layer			D	S
x_1	x_2	x_3	x_1	x_2	x_3		
benzene (1) + cyclohexane (2) + [C ₂ mim][Ac] (3)							
0.085	0.915	0.000	0.036	0.021	0.943	0.42	18.34
0.175	0.825	0.000	0.085	0.069	0.847	0.48	10.33
0.264	0.736	0.000	0.140	0.040	0.820	0.53	10.15
0.360	0.640	0.000	0.180	0.034	0.776	0.50	9.56
0.462	0.538	0.000	0.212	0.033	0.755	0.46	7.44
0.561	0.439	0.000	0.218	0.040	0.742	0.39	4.26
benzene (1) + cyclohexane (2) + [C ₂ mim][N(CN) ₂] (3)							
0.073	0.927	0.000	0.049	0.020	0.932	0.67	31.85
0.158	0.842	0.000	0.095	0.018	0.886	0.61	27.94
0.247	0.753	0.000	0.141	0.016	0.844	0.57	27.48
0.339	0.661	0.000	0.184	0.015	0.801	0.54	23.51
0.426	0.574	0.000	0.232	0.016	0.752	0.54	19.23
0.522	0.478	0.000	0.275	0.015	0.710	0.53	16.84
benzene (1) + cyclohexane (2) + [C ₂ mim][SCN] (3)							
0.082	0.918	0.000	0.038	0.010	0.953	0.46	44.71
0.168	0.832	0.000	0.077	0.012	0.910	0.46	31.18
0.259	0.741	0.000	0.114	0.010	0.876	0.44	32.73
0.351	0.649	0.000	0.149	0.009	0.842	0.42	30.94
0.450	0.550	0.000	0.184	0.008	0.808	0.41	26.53
0.553	0.447	0.000	0.222	0.007	0.771	0.40	25.22
benzene (1) + cyclohexane (2) + [C ₂ mim][Tf ₂ N] (3)							
0.078	0.922	0.000	0.095	0.059	0.846	1.22	19.09
0.157	0.843	0.000	0.185	0.058	0.757	1.18	17.14
0.243	0.757	0.000	0.259	0.057	0.685	1.07	14.28
0.332	0.668	0.000	0.331	0.054	0.615	1.00	12.36
0.417	0.583	0.000	0.396	0.054	0.549	0.95	10.24
0.518	0.482	0.000	0.459	0.054	0.487	0.89	7.91

Table 6

Parameters of Othmer-Tobias and Hand correlation for each ternary system and the values of regression coefficient R^2 .

Ternary system	Othmer-Tobias			Hand		
	a	b	R^2	c	d	R^2
[C ₂ mim][Ac]	1.400	2.297	0.878	1.175	2.142	0.933
[C ₂ mim][N(CN) ₂]	1.534	2.624	0.999	1.306	2.261	0.998
[C ₂ mim][SCN]	1.481	2.977	0.993	1.320	2.681	0.991
[C ₂ mim][Tf ₂ N]	1.458	2.239	1.000	1.190	1.906	0.996

molar composition of the tie lines generated by COSMO-RS and the corresponding RMSD values are respectively tabulated in Table S1 and S2 of the Supporting Information. The average RMSD value between the experimental and COSMO-RS was 4.88%. Considering that COSMO-RS is a fast and an a priori method, this value supports the advantage of COSMO-RS as an assisting tool for thermodynamic properties calculations.

4.6. Comparison of selected ILs with other solvents

The efficiency of the extraction process can be evaluated from the distribution ratio of solute (benzene in this case) and the selectivity of the solvent. Figs. 14, 15, and 12 compare the values from this work with some of the previous reports that used different types of extracting solvent, namely, organic solvents [34, 35], ILs [11, 15, 36–39] and DESs [17]. Sulfolane was included as a benchmark to represent the performance of the common organic solvents used in industry. In addition, the organic solvents with high selectivity (ethylene glycol (EG)) and distribution ratio (dimethylformamide (DMF)) were also included for comparison [34]. At low concentrations of benzene in the raffinate phase ($x_B < 0.3$), the distribution ratio for all ILs were higher than that of sulfolane. Furthermore, all ILs in this study also displayed higher selectivity than that of sulfolane, confirming the potential of ILs as alternatives solvents.

It was also observed that the extraction performances of the ILs were higher than that of the DESs. These results indicate the advantages and drawbacks of each extracting solvent in the extractive separation of benzene and cyclohexane. Although the organic solvents are cheaper with moderate performance, they are limited by their high volatility and complex process. While ILs have higher extractive performance than organic solvents and DESs, their use is limited because they are expensive and possibly toxic [40]. On the other hand, although the DESs are cheaper, can be easily synthesized, and environmentally friendly, their extractive performance is rather low. Nevertheless, as shown in

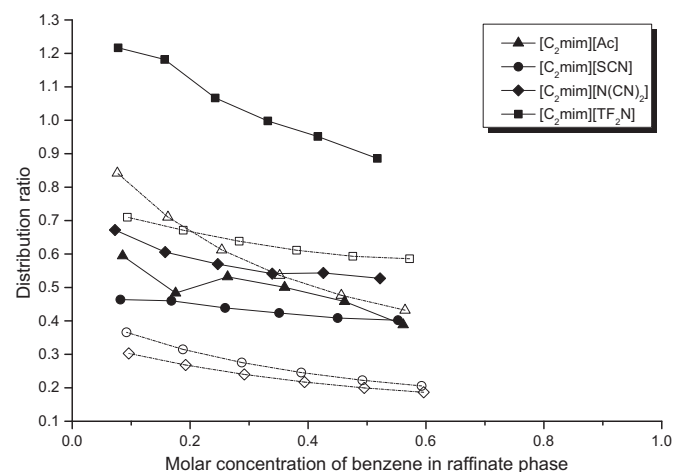


Fig. 12. Distribution ratio of benzene versus mole fraction of benzene in raffinate phase for five ternary systems in this study. The solid and dashed lines indicate experimental and COSMO-RS data, respectively.

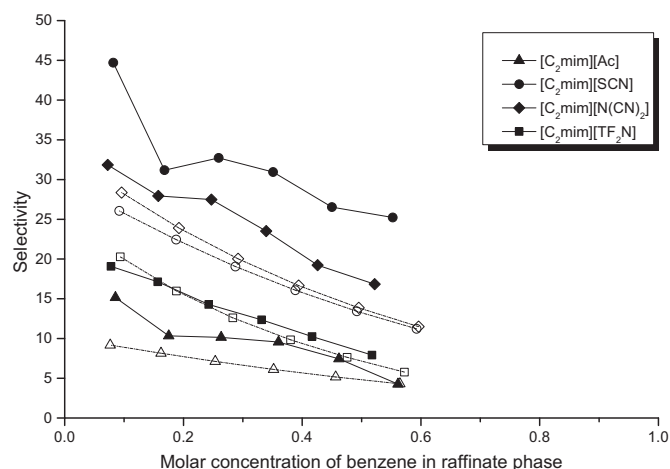


Fig. 13. Selectivity of ILs versus mole fraction of benzene in raffinate phase for five ternary systems in this study. The solid and dashed lines indicate experimental and COSMO-RS data, respectively.

our previous study, DESs can still be used because of efficient extraction with multiple stages and easy regeneration [17].

It is noteworthy that the three ILs, namely, [C₄mim][TF₂N] [36], [C₄mim][PF₆] [37], and [C₄mim][AlCl₄] [41], showed exceptionally higher distribution ratios than our work. However, the use of each of these ILs as the extracting solvent was associated with significant drawbacks. Although the extraction using [C₄mim][TF₂N] showed the highest distribution ratio, its selectivity was relatively much lower than that of the other ILs. For the extraction using [C_nmim][PF₆] (*n* = 4, 5, 6) [37], the high distribution ratio was mitigated by the cost of operation because it is a highly expensive type of IL. The IL [C₄mim][AlCl₄] is reportedly unstable in humid air, which makes process handling difficult.

4.7. NRTL correlation

In LLE calculations, an isothermal liquid–liquid flash is solved at a given temperature and pressure to obtain the phase compositions. The flash calculation consists of the Eqs. 10, 11, and 12.

Material Balance:

$$x_i - (1 - \omega)x_i^{L1} - \omega x_i^{L2} = 0, \quad i = 1, N_c \quad (10)$$

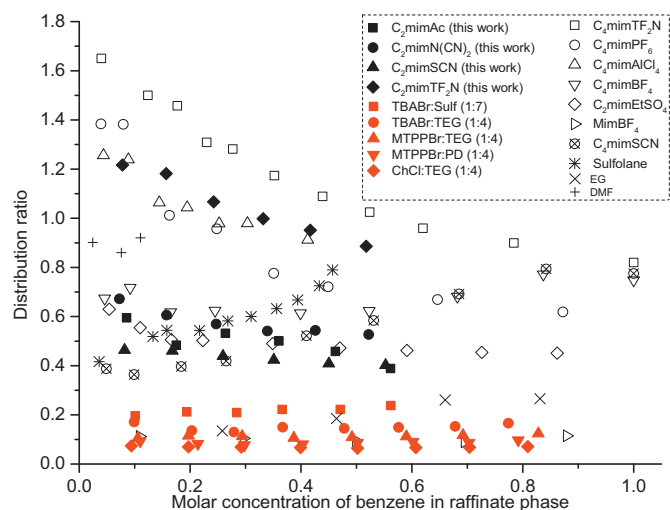


Fig. 14. Distribution ratio of organic solvents, ILs and DESs for the extractive separation of benzene and cyclohexane.

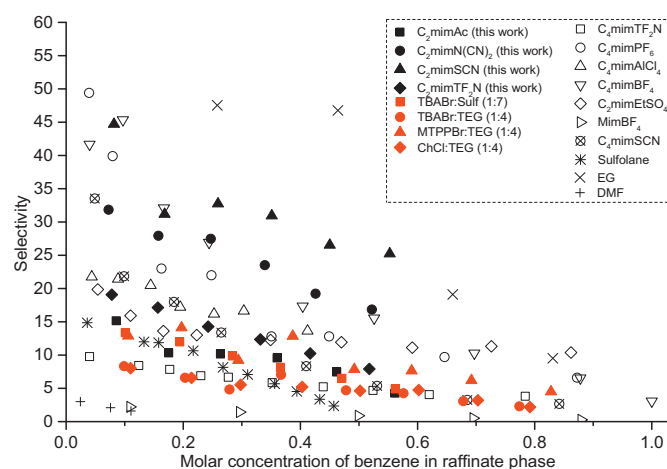


Fig. 15. Selectivity of organic solvents, ILs and DESs for the extractive separation of benzene and cyclohexane.

Equilibrium Equation:

$$x_i^{L1} \gamma_i^{L1} - x_i^{L2} \gamma_i^{L2} = 0, \quad i = 1, N_c \quad (11)$$

Equation of Summation:

$$\sum_i x_i^{L1} - \sum_i x_i^{L2} = 0 \quad (12)$$

Here, ω is the liquid–liquid splitting ratio, x_i refers to the amount of component i in the mixture, x_i^{L1} stands for the amount of component i in liquid phase $L1$, x_i^{L2} refers to the amount of component i in liquid phase $L2$, and N_c is the number of constituents in the liquid phase. The parameters γ_i^{L1} and γ_i^{L2} are the activity coefficients of component i in $L1$ and $L2$, respectively.

In this work, the activity coefficients were evaluated using the non-random two-liquid (NRTL) model [42]. This model was found to be useful for correlating the experimental LLE data of systems containing ILs without requiring any special modification. For a multicomponent system, the activity coefficient of component i is given by the general expression, shown in Eq. 13:

$$\ln \gamma_i = \frac{\sum_j \tau_{ji} G_{ji} x_j}{\sum_j G_{ji} x_j} + \sum_j \frac{G_{ji} x_j}{\sum_k G_{kj} x_k} \left(\tau_{ij} - \frac{\sum_k \tau_{kj} G_{kj} x_k}{\sum_k G_{kj} x_k} \right) \quad (13)$$

with: $\ln G_{ij} = -\alpha_{ij} \tau_{ij}$, $\alpha_{ij} = \alpha_{ji}$ and $\tau_{ii} = 0$, where τ_{ij} and τ_{ji} are binary interaction parameters and α_{ij} is the non-randomness parameter.

The binary interaction parameters τ_{ij} and τ_{ji} were estimated by minimizing the RMSD between calculated and experimental solubilities of each constituent in each phase using the Simulis® software package [43]. The parameter α_{ij} in the NRTL model measures the non-randomness in the mixture; i.e. the mixture is said to be completely random when α_{ij} is zero. In this work, α_{ij} was fixed equal to 0.20 for all binary combinations. The choice of $\alpha_{ij} = 0.2$ is made because of the successful fitting of the NRTL model to the ternary LLE data for systems containing ILs and DESs, as reported in our previous works [17, 44].

The RMSD values between the experimental and NRTL calculations for both ternary systems were <2%, which indicates that NRTL correlation represents the experimental data very well. This excellent fitting is also obvious from Figs. 8–11. Table 7 shows the values of the NRTL binary interaction parameters regressed for each ternary system. In order to conserve the coherence between this work and the previous report applied to DESs for the same binary mixture, the binary interaction parameters between benzene and cyclohexane, despite the nature of the IL used, were taken from our previous work [17] without any adjustment.

Table 7

Binary interaction parameters in NRTL regression for the ternary systems in this study with RMSD between experimental and calculated data.

Ternary system	<i>i</i>	<i>j</i>	τ_{ij}	τ_{ji}	RMSD (%)
	Benzene	Cyclohexane	–189.57	453.65	
Benzene + cyclohexane + [C ₂ mim][Ac]	Benzene	[C ₂ mim][Ac]	3924.03	64.56	1.174
	Cyclohexane	[C ₂ mim][Ac]	1780.45	535.69	
Benzene + cyclohexane + [C ₂ mim][N(CN) ₂]	Benzene	[C ₂ mim][N(CN) ₂]	5016.78	148.12	0.881
	Cyclohexane	[C ₂ mim][N(CN) ₂]	2108.46	1340.29	
Benzene + cyclohexane + [C ₂ mim][SCN]	Benzene	[C ₂ mim][SCN]	4424.45	159.42	0.614
	Cyclohexane	[C ₂ mim][SCN]	1911.05	1655.71	
Benzene + cyclohexane + [C ₂ mim][Tf ₂ N]	Benzene	[C ₂ mim][Tf ₂ N]	7289.11	110.31	1.174
	Cyclohexane	[C ₂ mim][Tf ₂ N]	1878.57	532.41	

Finally, in order to confirm the consistency of the binary interaction parameters obtained in this work, we have used the topological analysis of liquid-liquid equilibrium correlations designed recently by Marcilla et al. [45] and made available online as a Matlab toolbox. According to this test, the binary interaction parameters (reported in Table 7) yield to very good agreement with experimental tie-lines and at the same time satisfy the Gibbs stability criteria and are coherent with the phase occurrence obtained when mixing the different binaries. In fact, according to our lab experiments, when mixing benzene (1) with cyclohexane (2) only one homogeneous phase occurs. However, when mixing either benzene (1) or cyclohexane (2) with one of the four ILs (3) two layers appear. The same results are obtained when calculating the binary Gibbs mixing energy (provided in the supplementary Information in Figs. S1 to S4). The miscibility boundary in both cases confirms this results with one homogeneous region for 1–2 and a single liquid-liquid region for the 1–3 and (2–3).

5. Conclusion

In this work, COSMO-RS was used to screen 208 ILs for the separation of benzene from its mixture with cyclohexane. Four of the top-screened ILs were selected for experimental validation in the liquid-liquid extraction process. In addition to being commercially available at affordable prices, all ILs showed good performance in the extraction of benzene from the benzene-cyclohexane mixture. The experimental results confirmed that COSMO-RS is a fast, reliable, and useful tool to qualitatively screen the performance of ILs in the extractive separation of benzene and cyclohexane. In addition, there was no IL present in the cyclohexane layer and the concentration of cyclohexane in the IL layer was very low and the performance of the ILs selected in this work was found to be better than that of classical organic solvents such as ethylene glycol, dimethylformamide and sulfolane, confirming the potential of ILs as alternatives solvents. Moreover, the NRTL model was successfully employed to correlate the tie-lines in all ternary diagrams.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.molliq.2018.06.034>.

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